

MEWS Development Program

Design Input Document (DID) — Sample Portfolio Artifact

1. Purpose

This sample Design Input Document translates system and software requirements into design inputs that can guide architecture, hardware/software interfaces, implementation, verification, and validation. It is intentionally simplified for portfolio use.

2. Design Input Categories

Category	Design Input
System Architecture	The solution shall support an integrated wearable, gateway/connectivity, cloud, backend, and monitoring-application architecture.
Device Interface	The design shall define the interface between wearable-device data acquisition and the connectivity layer.
Gateway Interface	The gateway design shall provide the connectivity path required to transfer device observations toward cloud services.
Data Model	The design shall represent patient / encounter context, observations, device information, configuration, and relevant operational records.
Application	The application shall provide monitoring views and workflows required by authorized users.
Cloud / Backend	Backend services shall provide APIs and processing required to receive, validate, store, and expose supported monitoring information.
Security	The design shall include authentication, authorization, secure communication, and protection of sensitive healthcare information.
Quality / Traceability	Design inputs shall remain traceable to approved requirements and downstream verification / validation evidence.
Manufacturing / Serviceability	Physical design inputs shall consider assembly, enclosure, device configuration, and readiness for downstream production activities.

3. Physical Design References

The supplied images provide visual reference for the wearable form factor, prototype configuration, and gateway enclosure. They are referenced here as design-context evidence rather than as controlled engineering drawings.



Figure 1 — Wearable form-factor reference

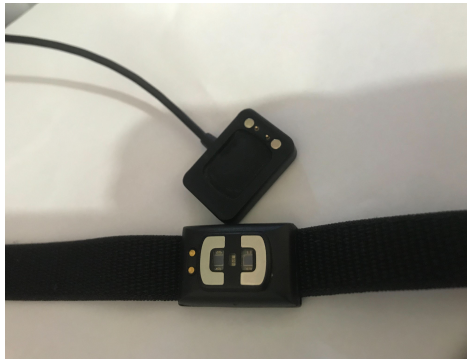


Figure 2 — Wearable prototype reference



Figure 3 — Wearable-on-user reference



Figure 4 — Gateway enclosure reference

4. Verification Relationship

Design outputs should be reviewed against the approved design inputs. Verification evidence should demonstrate that the implemented design satisfies its specified inputs; validation should demonstrate suitability for the intended use and workflow.

5. Portfolio Boundary

This sample does not reproduce proprietary algorithms, controlled engineering drawings, component part numbers, credentials, patient information, regulatory submission records, or confidential source material.